

Identification and Date-Stamping of Bubbles and Detecting Co-Explosive Behaviour: Energy and Non-Energy Commodity Market (1970 – 2020)

University of Nottingham

School of Economics

MSc Economics and Econometrics (2020 – 2021)

Supervisor: Professor Steve Leybourne

Student ID: 20311274

This Dissertation is presented in part fulfilment of the requirement for the completion of an MSc in the School of Economics, University of Nottingham. The work is the sole responsibility of the candidate.

Abstract

In this paper, we study the explosive price behaviour in the energy and non-energy sector using 12 price series for the last 50 years (1970 – 2020). The right-tailed test suggested by Phillips, et al. (2015) is used to test the explosive autoregression in individual price series and date-stamp our estimated origination and termination of episodes of exuberance. The recently developed variant of the stationarity test of Kwiatkowski, et al. (1992) by Evripidou, et al. (2021) is then employed which uses a wild-bootstrap procedure that generates critical values that are robust to heteroskedasticity, on these explosive series to test for co-explosive behaviour between these series. All price series are found to exhibit explosive autoregressive behaviour. From the 66 possible pairs of price series, 14 co-bubbling pairs of series are found with five pairs showing contemporaneous co-explosive behaviour. The maximum time it takes to migrate from one series to the other is found to be 22 months. Factors influencing the commodity markets and major events that took place in the last 50 years are studied to understand the individual explosiveness and co-bubbling.

Keywords: Right-tailed, explosive autoregression, co-bubble, wild bootstrap.

Word Count: 12,031.

I am grateful to my supervisor Professor Steve Leybourne for providing the Gauss Code for the variant of the KPSS test and also for his guidance.

Contents

1	Introduction	4
2	Literature Review	7
2.1	Bubble in energy market	7
2.2	Co-movement between energy prices and non-energy prices	8
2.3	Bubble testing methods	9
3	Methodology	12
3.1	Tests for explosive autoregression in the individual price series	12
3.2	Bubble Migration Testing	17
4	Data Description	22
5	Results	25
5.1	Explosive autoregression in the individual series	25
5.2	Test for co-explosive behaviour	29
5.3	Policy Implications	33
6	Concluding Remarks	35
	Bibliography	37
	Appendices	40
	Appendix 1 – Right-tail ADF statistics	40
	Appendix 2 – Right-tail ADF statistics (sub-sample)	41
	Appendix 3 – Date-stamps	41
	Appendix 4 – Results of Johansen cointegration test statistics	42
	Appendix 5 – Results ADSR for co-explosive behaviour	43

Chapter 1

Introduction

The U.S stock market experienced spectacular increases in all the major indices including Nasdaq index during the 1990s which was mainly led by the internet sector and DotCom stocks (Phillips et al., 2011). There has been a lot of debate about what caused this unusual surge and the fall in prices, whether there were bubbles¹ or not, and if there were bubbles whether the bubbles were behavioural or rational. Then during the boom and bust of the 2006 – 09 Global Financial Crisis (GFC), the world experienced a significant surge in commodity prices which is the focus of this paper; energy and non-energy sector. We will be studying the price behaviour of energy and non-energy sector using data from September 1970 – August 2020 for a selected number of price series from these two sectors.

Some economists and econometric literature attribute the sudden changes in the prices of commodities to episodes of financial (speculative) bubbles while some link it to particular non-financial events. Most speculative bubbles in history have been studied and researched ex post, using econometric techniques and procedures to determine whether it is the self-fulfilling speculation or market fundamentals² that lead to these bubble or price patterns (Saitone and Sexton, 2007). The former Federal Reserve Board Chairman, Alan Greenspan used the phrase “irrational exuberance” to explain the herd stock market behaviour. This remark has been very influential in providing perceptions about the financial markets and herd behaviour (Phillips et al., 2011).

The catastrophic Global Financial Crisis (GFC) of 2006 – 09 took place just after a decade of his speech making it one of the most debatable topics in the literature. It became interesting for researchers and extremely important for policymakers because the effects of such bubbles not only affect the financial sector per se, but also other sectors such as energy and non-energy sector. For this reason, it is imperative that they know the causes and effects of such bubbles. For example, in the case of GFC, one view of research points to the massive global savings

¹Garber (2000) defines a “bubble” as the unexplained part of the asset price movement that could not be explained by the fundamentals.

²Fundamentals are a collection of variables that we believe are causing or contributing to the asset price movements.

surplus and huge capital inflows into the United States and Europe arising from the Asian financial crisis as a key determinant of the 2006 – 2009 crisis (Roy and Kemme, 2020). Another view of research contends that the GFC was primarily caused by prospects of unending asset prices and due to the increasing regulatory imprudence caused by great moderation, thereby concluding that the rise in the asset prices happened due to lax regulations, predatory lending and extreme risk taking by financial establishments.

The dramatic GFC-led international commodity price surges we mentioned earlier was the longest and the broadest price surge since World War II period, which enlivened the debate over the determinants of the agricultural commodities and the factors contributing to sudden surges. This event of 2006 – 09 caused much interest in the policy actions both at national and global level (Gutierrez, 2013). While several potential determining factors have been identified in the literature, there has been no agreement on the relative weight that should be credited to these factors. Authors such as Braun (2007) and Dewbre, et al. (2008) suggest that biofuel programmes in the US and in the Europe might be behind the increasing food prices as their programmes provide much subsidy for biofuels, which results in greater usage in vegetables and crops, ultimately resulting in the increased commodity prices. Some authors point to the link between energy prices and non-energy prices as the main determinant for the commodity price movements especially in the food and agricultural markets. One such author is Baffes & Haniotis (2010) according to whom the cost of input and cost of transportation can greatly affect the food prices via energy prices. Therefore, it is important that we identify the causes and effects of these sudden surges in energy and non-energy sectors and whether these surges migrate from one sector to the other or within the same sector or not.

Several econometric techniques have been developed to detect bubbles in assets in the context of the present value of dividends model (Gordon and Shapiro, 1956). It uses stock price as an application and suggests that the stock price can be separated into two components, a “market fundamental” component which is the discounted value of expected future dividends and a “bubble” component which is non-stationary. The latter component is exploited by some of the traditional econometric tests of bubble such as Variance Bounds tests, West’s two-step tests and integration/cointegration based tests (Gürkaynak, 2008).

Variance Bound tests of Shiller (1981) and LeRoy & Porter (1981) although not initially designed to detect bubbles, is considered to be one of the first bubble detection procedures to be developed. While Shiller’s tests only generate points estimates of variances, which made testing statistical significance impossible, LeRoy and Porter’s test, essentially a Vector autoregression (VAR) test (of the market fundamental prices), constructs estimate of variances with standard errors by treating stock prices and dividends as bivariate processes.

Another bubble testing method is West’s test (West, 1987) which explicitly puts bubbles as an alternative hypothesis. But this test only detects bubbles and not the origination or the termination dates of the bubbles. Both Variance Bound tests and West’s test provide little detail about the structure of the bubbles. Campbell & Shiller (1987) and Diba & Grossman (1988b)

introduced the right tailed unit root test and the cointegration test which became one of the most commonly used bubble detecting methods in the literature, and these have been widely used to detect housing price bubbles and stock market bubbles in the last 10 years.

However, due to the many limitations identified by Evans (1991) on these tests, we will be using more modern econometric techniques, namely Phillips, et al. (2015b) (hereafter PSY) to test for explosive autoregression in individual price series of energy and non-energy sector and the recently developed variant of the stationarity test of Kwiatkowski, et al. (1992) by Evripidou, et al. (2021) (hereafter ADSR) to test for co-explosive behaviour in and between those two sectors.

This remainder of this paper is structured as follows. In Section 2, we briefly analyse the energy market and non-energy market and bubble testing methods. In Section 3, we describe the methodologies we use for testing explosive autoregression in individual price series, co-bubble testing and the wild bootstrap procedure. In Section 4, we describe the data we are using for this study. In Section 5, we present our findings, explain the results and policy implications and Section 6 concludes the paper.

Chapter 2

Literature Review

2.1 Bubble in energy market

Heavy disruptions take place in the global oil market with considerable regularity. The Gulf War 1991/90 led oil price hike and the July 2018 episodes of oil price reaching 140 USD per barrel are notable examples of oil crises (Gronwald, 2016). Each of these crises triggered much debate in the literature and in the public and considerable effort has been made to understand the behaviour and macroeconomic causes and effects of oil prices. These oil price disruptions were frequent and price fluctuations were dramatic as well. The prices seemed to be moving horizontally prior to 1986, with only prices dramatically shifting in 1973/1974, 1979/1980 and in 1986. This horizontal movement continued into the 21st century with a higher volatility, with a steady growth to observe a record high in 2008. After the explosiveness in 2007/2008 it continued the increasing trend before it collapsed before the dramatic collapse in 2014. These sudden movements are called jumps and major price changes are called structural breaks. However, exceptions were observed in 1990/1991 and in 2018 in both cases the steep increases were followed by a subsequent collapse. It also took a considerable amount of time to reach back to its pre-collapse period levels; more than three months in the case of the 1990/1991 collapse and even longer for 2008. This behaviour cannot be described as a structural break as these are identified as isolated jumps or only an increase in the volatility (Gronwald, 2016).

The author employed the ‘temporary explosiveness’ concept to capture this interesting behaviour by using a forward recursive application of augmented Dickey-Fuller (ADF) unit root test where the null hypothesis of a unit root is tested against the alternative of explosiveness. It found evidence of explosiveness for the periods 1990 – 1991, 2005 – 2006 and 2007 – 2008 – this is a major contribution to empirical literature on oil price behaviour as the unique subsequent collapses in each period are conceptually different from the conventional oil price behaviour.

Some econometric literature such Kilian & Murphy (2014), using structural VAR models, show that these episodes are mainly caused by fundamental factors such as global business

cycle led oil consumption and that speculative trading has a minimal role. Moreover, publicly popular claims that the 2004 – 2008 oil price hikes were caused by the ‘financialisation’¹ of the futures oil market were also debunked empirically and using Granger Causality and long-horizon regression tests by Irwin & Sanders (2012). Therefore, it can be empirically demonstrated that these extreme oil price movements have fundamental explanations such as the relationship between crude oil supply and its demand and the low-price elasticities of oil demand and supply. However, some econometric literature does not sufficiently acknowledge this finding that they are caused by fundamental causes. Papers such as Phillips, et al. (2011) and Brooks, et al. (2015) deal with speculative bubbles by employing modern econometric techniques, which will be explained in this paper later on.

2.2 Co-movement between energy prices and non-energy prices

The non-energy market we refer to in this paper is the agricultural and food commodity market. The interlinkage between the energy and non-energy market is typically associated with transportation and production costs. However, in recent times, this interlinkage has become stronger due to the development of biofuel industry and increase in demand for agricultural products to produce ethanol and biodiesel (De Nicola et al., 2016). Many economists also attribute the price changes in the agricultural products to the role of exchange rate, monetary policy and its macroeconomic transmission mechanisms (Saghaian, 2010). However, as mentioned before the nature of the relationships between ethanol, oil and commodity prices have become an interesting question among the agricultural economists. The integration of energy and agricultural markets has the potential to lead to more volatility in their prices, especially in the agricultural prices. In the paper that examined 35 globally traded commodities for the period 1960 – 2005, for effects of crude oil, it found oil price changes had the most potential and highest pass-through (elasticity) to fertilisers and food commodities among the non-energy market (Baffes, 2007). It places the pass-through of crude price oil changes to agriculture at 0.17 and the overall non-energy commodity index at 0.16. Peñaranda & Micola (2009) attempted to systematically analyse the structural connection between crude oil and agricultural prices through the emergence of biofuel. This was the first such known study of this relationship. Their research shows that between 2006 – 2008, on Chicardgo Board of Trade, the price of corn increased by 60%, soyabeans by 76% wheat by 54% and rice by 104% which then went down equally by similar percentages. At the same time, the price of crude oil went up by almost 100% from 70 USD to 140 USD per barrel and then went down to 40 USD per barrel. Some economists described it as a speculative bubble.

However, it is interesting to note that these events coincided with the adoption of ethanol-

¹This is also called the ‘Masters Hypothesis’ which claims that long-only index investments were a major cause of the 2007 – 2008 commodity futures price hikes and more particularly in energy futures (Irwin & Sanders, 2012).

based biofuels in the U.S transportation sector. A generally held view is that the U.S biofuel industry's emergence was due to the combination of the 2005 Energy Policy Act and the extremely high oil prices (Peñaranda and Micola, 2009).

2.3 Bubble testing methods

There has been a lot of attention in the theoretical and empirical research on why asset prices show excess volatility compared to the efficient market hypotheses or models. Focusing on commodity prices, the literature widely accepts that commodity prices might deviate from its instinct values (which comes from the fundamentals) due to 'speculative bubbles' or 'information bubbles' (Gutierrez, 2013). Kleidon (1986) suggests that the excess volatility in asset prices might be caused by a fundamental model misspecification or due to investor irrationality rather than due to the presence of a bubble. Brooks & Katsaris (2003) also conclude their paper stating that divergence from economic fundamentals might be caused by investor irrationality, as suggested by Kleidon (1986), and that the efficient market models that are based on arbitrage conditions might be unsuitable to explain market outlooks and dynamic forces.

One of the most important strands of literature that has been developed to explain the divergence to asset prices from their fundamentals comes from the theory of "rational bubbles" (Kortian, 1995). This theory was also the first attempt in the academic literature to rigorously challenge the key propositions of Efficient Market Hypothesis. It explains that "rational deviations" or "rational bubbles" are still possible even with rational expectations and behaviour. The same argument has been suggested by authors such as Flood & Garber (1980) and Blanchard (1979).

The Efficient Market Hypothesis or Present Value Financial Theory is used to define the fundamental value of the asset price. Gutierrez (2013) applies this theory to agricultural commodity markets to. In this model, the commodity price P_t is the sum of the expected future price P_{t+1} and future payoffs ψ_{t+1} which are both discounted at a constant rate R . This highly reduced form of dynamic supply and demand model is represented by the following equation

$$P_t = \frac{E_t(P_{t+1} + \psi_{t+1})}{1 + R}, \quad (2.1)$$

where ψ is the convenience yield² for a storable commodity. It is the benefit that accrues to the holders of inventory for the possibility of making them available in periods of scarcity. This equation is used to explain price exuberance via a log-linear framework suggested by Campbell, et al., (1997). The relationship between the variables in the equation and possible bubbles will be tractable under empirically plausible assumptions using such a log-linear approximation. Campbell, et al., (1997) is followed by taking log-linear approximation of equation (1) to derive

²Convenience yield is the cost of insurance, storage and finance cost which are incurred in order to store a commodity. This explains why the spot prices are usually higher than future prices (Kaldor, 1939).

the following result through recursive substitution.

$$p_t = p_t^f + b_t, \quad (2.2)$$

where p_t is the log of commodity price, p_t^f is the fundamental price and b_t is the bubble variable. Here p_t is split into a fundamental component and a bubble component as mentioned in the beginning. This can then be written as

$$p_t^f = \frac{\kappa - \gamma}{1 - \rho} + (1 - \rho) \sum_{i=0}^{\infty} \rho^i E_t \psi_{t+1+i} \quad (2.3)$$

$$b_t = \lim_{i \rightarrow \infty} \rho^i E_t p_{t+i}, \quad (2.4)$$

where $p_t = \ln(P_t)$, $\psi_t = \ln(\psi_t)$, $\gamma = \ln(1 + R)$, $\rho = 1/(1 + \exp(\psi - \rho))$, with $\psi - \rho$ being the average long convenience yield – price ratio $0 < \rho < 1$, and

$$\kappa = -\ln(\rho) - (1 - \rho) \ln\left(\frac{1}{\rho} - 1\right) \quad (2.5)$$

According to Phillips, et al., (2011), while log-linear approximations like the above about sample mean are commonly used in both empirical and theoretical work, they are less efficient and less satisfactory in non-stationarity contexts; meaning when sample mean do not converge to population constants. For this reason, in their paper on stock prices, they used their series in both log levels and in levels, and it generated similar results for both cases. The equation (2) makes it clear that when $b_t = 0$, $b_t = \lim_{i \rightarrow \infty} \rho^i E_t p_{t+i} = 0$ (transversality condition is imposed), then p_t^f determines p_t ; meaning discounted expected ψ determines p_t (refer equation 3). However, if $b_t \neq 0$, then there exists infinite number of solutions to equation 1 and any or all solutions can be stated as equation 2. Authors such as Campbell, et al., (1997) call b_t a ‘rational bubble’ as the the presense of b_t is consistent with constant expected returns and rational expectations. Gutierrez (2013) and Phillips, et al. (2011) describe the rational bubble in the following way.

$$b_t = \frac{1}{\rho} b_{t-1} + \varepsilon_t \equiv (1 + g) b_{t-1} + \varepsilon_t \quad (2.6)$$

where $g = \left(\frac{1}{\rho}\right) - 1 = \exp(\psi - \rho) > 0$ is the growth rate of the natural log of bubble b_t and ε_t is the random error. Thus, regardless of whether the convenience yield ψ_t is stationary or not, the bubble b_t and the price p_t will be explosive as $\exp(\psi - \rho) > 0$. If the changes in price p_t is stationary, then the presence of any explosive rational bubbles is removed, otherwise the rational bubble hypothesis we mentioned earlier holds. This is also the method Diba & Grossman (1988a) used to test for explosive rational bubbles.

The early bubble testing methods that were briefly explained in the introduction, Variance Bound tests (Shiller, 1981), West’s two-step test (West, 1987) and the right-tailed unit root

test and cointegration (Diba & Grossman, 1988a) also use the present value model under the assumption of rational bubbles. However, Evans (1991) showed flaws in these standard bubble detecting unit root testing procedures by explaining that the data might appear stationary or as unit root throughout the sample period if the sample data include periodically collapsing bubbles.

Chapter 3

Methodology

3.1 Tests for explosive autoregression in the individual price series

Following PSY, three tests will be conducted based on the following ADF regression.

$$\Delta y_t = \alpha_{r_1, r_2} + \beta_{r_1, r_2} Y_{t-1} + \sum_{i=1}^k \delta_{r_1, r_2}^i \Delta y_{t-1} + \varepsilon_t \quad (3.1)$$

where Δ is the first difference operator, y_t is the time series of interest at time t , k is the number of dependent variables. r_1 and r_2 are the starting and ending points for the purpose of estimation, respectively. r_1 and r_2 are written as fractions of the total number of periods T . Therefore, $r_2 = r_1 + r_w$ (r_w is the size of the window of the regression which is also expressed as a fraction of T). Finally, the number of observations to estimate equation 7 is $T_w = [Tr_w]$ where $[.]$ is the floor function – it gives the integer part of the argument. ε_t is the error term. The testing hypotheses are then given as follows:

$$\begin{aligned} H_0 : \beta_{r_1, r_2} &= 0 \quad (\text{unit root}) \\ H_1 : \beta_{r_1, r_2} &> 0 \quad (\text{explosive behaviour}) \end{aligned}$$

The ADF t-statistic that is required to test the null hypothesis in equation 7 will be denoted as $ADF_{r_1}^{r_2}$.

$$ADF_{r_1}^{r_2} = \beta_{r_1, r_2} / \text{s.e.}(\beta_{r_1, r_2}) \quad (3.2)$$

Two statistics explored by PSY will be calculated by applying the above setting.

Standard ADF Statistic: right-tailed ADF using the full range of observations available, denoted by ADF_0^1 by fixing $r_1 = 0$ and $r_2 = 1$ (this means $r_w = 1$). The limit distribution of ADF_0^1

under the H_0 is given by,

$$\frac{\int_0^1 W dW}{\left(\int_0^1 W^2\right)^{1/2}}, \quad (3.3)$$

where W is a Wiener process [note: $\tilde{W}(r) = W(r) - \frac{1}{r} \int_0^1 W$ is the demeaned Brownian motion]. ADF_0^1 will be compared with the critical value from its limit distribution to test for any bubbles. The H_1 is that the exuberance is there over the whole sample. This is due to the fact that a standard ADF test like this does not perform well in maintaining the consistency for regime changes as it can exhibit low power if there are episodes of boom-busts. This is shown diagrammatically below.

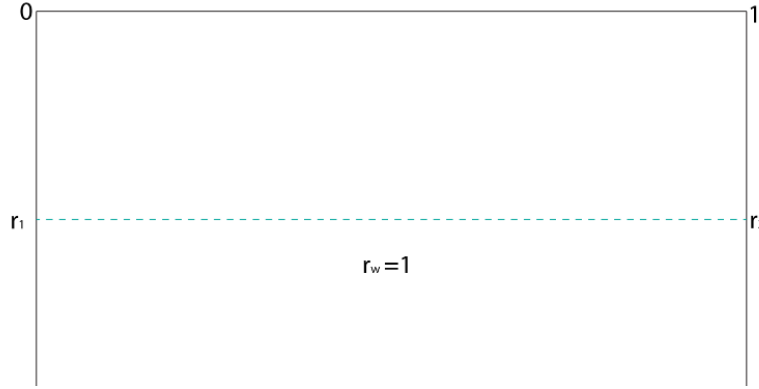


Figure 3.1: Sample sequences and window widths (full sample)

Supremum ADF Statistic: This is the test statistic Phillips, et al. (2011) proposed that is consistent with a single boom period. This involves estimating the equation 7 via a forward expanding sample. Here the difference is, the starting period of the sub-sample is fixed at $r_1 = 0$. At the same time, the end of the sub-sample (r_2) increases from r_0 (this is considered the minimum window size) to one (this is considered the whole sample period). Then a forward recursive estimation of equation 7 will yield a sequence of $ADF_0^{r_2}$ statistics. The supremum of this sequence is then defined as,

$$SADF(r_0) = \sup_{r_2 \in [r_0, 1]} ADF_0^{r_2} \quad (3.4)$$

with a limit distribution of the following.

$$\sup_{r_2 \in [r_0, 1]} \frac{\int_0^{r_2} W dW}{\left(\int_0^{r_2} W^2\right)^{1/2}} \quad (3.5)$$

If the supremum ADF exceeds the right-tail critical values, similar to the standard ADF test, the H_0 will be rejected. The fundamental difference between the standard ADF and the supremum ADF is that in the latter, the H_1 of SADF is that explosivity is present in only some parts of the sample whereas the standard test examines the whole sample period for the presence of

explosive dynamics. This is also shown diagrammatically below.

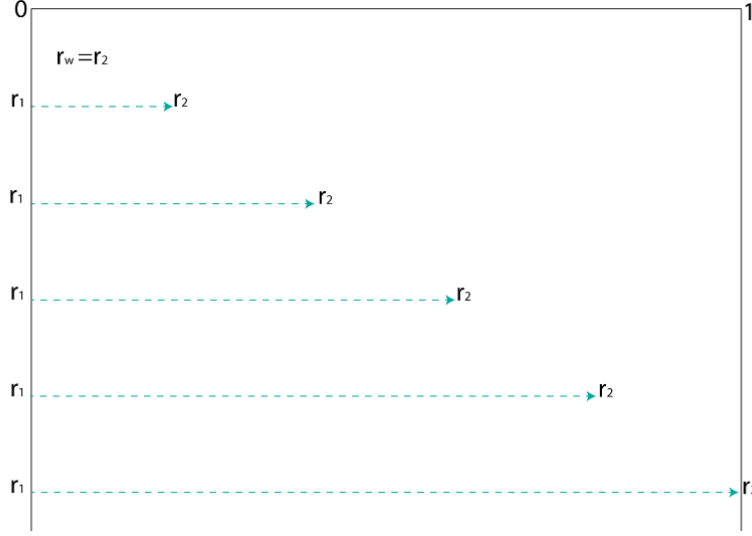


Figure 3.2: Sample sequences and window widths (recursive window)

However, if there are multiple bubbles, the recursive approach only provides a consistent estimate for the bubble starting and ending dates for the initial bubble, and not for the subsequent bubbles.

Generalised Supremum ADF (GSADF) Statistic

In order to overcome this issue, Phillips, et al. (2015) introduced a third statistic, the GSADF. As this is a more extensive approach where it uses more set of regressions, the first observation used for the purpose of estimation ranges from 0 to $r_2 - r_0$ and the last observation ranges from r_0 to 0. This has the same H_0 as SADF but it does cover more sub-samples. Given that there is a minimum window size required, r_0 equation 7 for all possible samples is estimated by allowing the starting point r_1 and the ending point r_2 to change thereby providing much flexibility. This extra flexibility on the windows yields substantial gains in terms of power and performance. This makes GSADF more appropriate and suited to detect presence of multiple regime changes compared to SADF. The GSADF statistics is described as,

$$GSADF(r_0) = \sup_{r_2 \in [r_0, 1], r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2} \quad (3.6)$$

with the following limit distribution under H_0 .

$$\sup_{r_2 \in [r_0, 1], r_1 \in [0, r_2 - r_0]} \left\{ \frac{\frac{1}{2} r_w [W(r_2)^2 - W(r_1)^2 - r_w] - \int_{r_1}^{r_2} W(r) dr [W(r_2) - W(r_1)]}{r_w^{1/2} \left\{ r_w \int_{r_1}^{r_2} W(r)^2 dr - \left[\int_{r_1}^{r_2} W(r) dr \right]^2 \right\}^{1/2}} \right\} \quad (3.7)$$

In equation 13, the size of the expanding window is given by $r_w = r_2 - r_1$. Here also the

rejection criteria for H_0 is the same as before i.e. the test statistic exceeding the right-tail critical value from limit distribution.

If the H_0 is rejected, then SADF and GSADF can provide a time line of the bubble episodes. Under the SADF methodology, the estimates of the starting date and the ending date can be obtained through the time series of the recursive, backward ADF test statistics. To do this, its statistics can be compared against the right-tailed critical values of the distribution of the standard Dickey-Fuller statistics and the origination (r_e) and termination (r_f) dates can be constructed from the following.

$$\begin{aligned}\hat{r}_e &= \inf_{r_2 \in [r_0, 1]} \{r_2 : ADF_0^{r_2} > cu_{r_2}^\alpha\} \\ \hat{r}_f &= \inf_{r_2 \in [\hat{r}_e, 1]} \{r_2 : ADF_0^{r_2} < cu_{r_2}^\alpha\}\end{aligned}$$

In the above, $cu_{r_2}^\alpha$ is the right-tailed critical value $(100 - \alpha)\%$ of the ADF statistic and α is the chosen significance level. The strategy for GSADF is also similar, but in this case, it is based on the sequence of backward SADF statistics which is defined below.

$$BSADF_{r_2}(r_0) = \sup_{r_1 \in [0, r_2 - r_1]} SADF_{r_1}^{r_2} \quad (3.8)$$

It estimates the origination and termination dates of the bubbles according to,

$$\begin{aligned}\hat{r}_e &= \inf_{r_2 \in [r_0, 1]} \{r_2 : BSADF_{r_2}(r_0) > scu_{r_2}^\alpha\} \\ \hat{r}_f &= \inf_{r_2 \in [\hat{r}_e, 1]} \{r_2 : BSADF_{r_2}(r_0) < cu_{r_2}^\alpha\}\end{aligned}$$

where $scu_{r_2}^\alpha$ is the right-tailed critical value $(100 - \alpha)\%$ of the SADF for $[r_2 T]$ observations. The consistency of the SADF strategy is established (in the case of a single bubble episode) in Phillips, et al. (2011) and GSADF for multiple explosive periods in Phillips, et al. (2015).

This is also shown diagrammatically below.

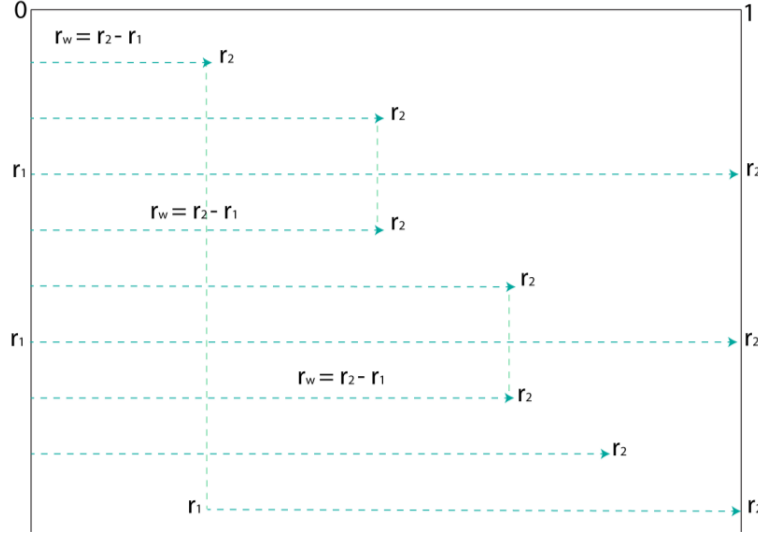


Figure 3.3: Sample sequences and window widths (recursive flexible window)

To make any inferences from these statistics, it requires right-tailed ADF, SADF and GSADF critical values generated from Monte Carlo Simulations. For the computational convenience, the tabulated critical values from Vasilopoulos, et al. (2020) will be used in this paper. Moreover, 499 bootstraps replications will also be generated for each price series with a maximum lag length of 6. All the right-tailed unit root testing procedures that were just detailed above require the application of iterative routines. These routines must fit the equation 7 repeatedly, on the subsamples of observations by solving a sequence which otherwise would be standard least-square (LS) problems. To compute these statistics, the STATA package will be used. It can be shown that by replacing the r_1 and r_2 with their discrete time analogues t_1 and t_2 , equation 7 can be re-written in the matrix form as below;

$$Y_{t_1:t_2} = X_{t_1:t_2} \beta_{t_1:t_2} + \varepsilon_{t_1:t_2}, \quad (3.9)$$

where

$$Y_{t_1:t_2} = \begin{bmatrix} \Delta y_{t_1} \\ \Delta y_{t_1+1} \\ \vdots \\ \Delta y_{t_2} \end{bmatrix}, \quad X_{t_1:t_2} = \begin{bmatrix} 1 & y_{t_1-1} & \Delta y_{t_1-1} & \cdots & \Delta y_{t_1-k} \\ 1 & y_{t_1} & \Delta y_{t_1} & \cdots & \Delta y_{t_1-k+1} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & y_{t_2-1} & \Delta y_{t_2-1} & \cdots & \Delta y_{t_2-k} \end{bmatrix},$$

$$\varepsilon_{t_1:t_2} = \begin{bmatrix} \varepsilon_{t_1} \\ \varepsilon_{t_1+1} \\ \vdots \\ \varepsilon_{t_2} \end{bmatrix}, \quad \beta_{t_1:t_2} = \begin{bmatrix} \alpha_{t_1:t_2} \\ \gamma_{t_1:t_2} \\ \psi_{t_1:t_2}^1 \\ \vdots \\ \psi_{t_1:t_2}^k \end{bmatrix}.$$

The solution to the LS problem for the subsample $t_1 : t_2$,

$$\underset{\beta_{t_1:t_2}}{\operatorname{argmin}} \|Y_{t_1:t_2} - X_{t_1:t_2} \beta_{t_1:t_2}\|^2, \quad (3.10)$$

is given by,

$$\beta_{t_1:t_2} = (X'_{t_1:t_2} X_{t_1:t_2})^{-1} X'_{t_1:t_2} Y_{t_1:t_2}. \quad (3.11)$$

3.2 Bubble Migration Testing

For bubble migration testing, we use the new variant of KPSS developed in ADSR which is based on a stationarity testing framework. It allows an explosive episode in one price series to lag or lead that in the other series by a certain number of periods, which in this case is considered as 12 months (plus or minus). The following generic data generating process for a generic series x_t is specified.

$$x_t = \mu_x + u_{x,t} \quad (3.12)$$

$$u_{x,t} = \begin{cases} u_{x,t-1} + \varepsilon_{x,t}, & t = 2, \dots, [\tau_{x,1}T], \\ (1 + \delta_{x,1})u_{x,t-1} + \varepsilon_{x,t}, & t = [\tau_{x,1}T] + 1, \dots, [\tau_{x,2}T], \\ (1 - \delta_{x,2})u_{x,t-1} + \varepsilon_{x,t}, & t = [\tau_{x,2}T] + 1, \dots, [\tau_{x,3}T], \\ u_{x,t-1} + \varepsilon_{x,t}, & t = [\tau_{x,3}T] + 1, \dots, T \end{cases}$$

where μ_x is a constant, $\tau_{x,1}$, $\tau_{x,2}$ and $\tau_{x,3}$ are sample fractions, $\delta_{x,1} > 0$ and $\delta_{x,2} > 0$. It is assumed that $u_{x,1} = o_p(T^{-1/2})$ and $\varepsilon_{x,t}$ is an error process with mean 0 and $I(0)$. This DGP imposes $I(1)$ on x_t up to $[\tau_{x,1}T]$, explosive AR process until $[\tau_{x,2}T]$, stationarity until $[\tau_{x,3}T]$ and $I(1)$ again until final regime. By changing $\tau_{x,1}$, $\tau_{x,2}$ and $\tau_{x,3}$, the DGP process allows for multiple model specifications, which are described below as in ADSR.

Model 1: $0 < \tau_{x,1} < \tau_{x,2} = 1$ (explosive regime emerges at $\tau_{x,1}$ and runs until sample end).

Model 2: $0 < \tau_{x,1} < \tau_{x,2} = \tau_{x,3} < 1$ (explosive regime terminates at $\tau_{x,2}$ and stays unit root until end of this sample).

Model 3: $0 < \tau_{x,1} < \tau_{x,2} < \tau_{x,3} = 1$ (a possible bubble crash – explosive regime terminates with an offsetting stationary collapse regime and continues until the end of sample).

Model 4: $0 < \tau_{x,1} < \tau_{x,2} < \tau_{x,3} < 1$ (the bubble crash or collapse is met with a unit root until the end of sample).

It will be assumed that y_t and x_t are two price series where x_t contains an explosive episode¹ that was generated from one of the above models.

The magnitude (speed) and duration of the explosive (bubble) period and the collapse is controlled by a combination of $\delta_{x,1}$ and $\tau_{x,2} - \tau_{x,1}$ and $\delta_{x,2}$ and $\tau_{x,3} - \tau_{x,2}$ respectively. In

¹For the purpose of simplicity, an explosive episode will be defined as an explosive regime which is followed by a stationary collapse regime.

addition to this, a new variable z_t will also be introduced which is the unobserved process which is also generated likewise (Model 1 – 4). But here x is replaced by z in equation 18. The corresponding model parameters are $\delta_{z,1,T}, \delta_{z,2,T}, c_{z,1}, c_{z,2}, \alpha_{z,1}$ and $\alpha_{z,2}$ where $\alpha_{z,1}, \alpha_{z,2} \in (0, 1)$. The explosive AR component $\delta_{x,1}$ is parameterised as $\delta_{x,1} = \delta_{x,1,T} = c_{x,1}T^{-\alpha_{x,1}}$ with $\alpha_{x,1} \in (0, 1)$ and stationary collapse AR component $\delta_{x,2} = \delta_{x,2,T} = c_{x,2}T^{-\alpha_{x,2}}$ with $\alpha_{x,2} \in (0, 1)$. After this, the DGP for y_t is defined as,

$$y_t = \mu_y + \beta_x x_{t-i} + \beta_z z_t + \varepsilon_{y,t}, \quad (3.13)$$

where $\varepsilon_{y,t}$ is a mean zero $I(0)$ error term. Correlation is allowed between $\varepsilon_{y,t}, \varepsilon_{x,t}$ and $\varepsilon_{z,t}$. In this equation, the explosiveness is either triggered by x_{t-i} (i.e. if $\beta_x > 0$ and $\beta_z = 0$) or z_t (i.e. if $\beta_z > 0$ and $\beta_x = 0$). It can also happen due to both, x_{t-i} and z_t (i.e. if $\beta_x > 0$ and $\beta_z > 0$). Interpreting equation 19, if $\beta_x > 0$ and $\beta_z = 0$, then y_t and x_{t-i} are ‘co-explosives’. This means when $\mu_y + \beta_x x_{t-i}$ is moved to LHS (linear combination), it becomes $I(0)$, implying that linear integration is stationary in all subsamples involved in y_t and x_{t-i} . Co-explosivity also implies cointegration in the $I(1)$ regimes of y_t and x_{t-i} in this case. Moreover, stationary linear combination in the explosive regimes is also implied in the co-explosive behaviour. The co-explosivity will be considered contemporaneous if $i = 0$. However, if $i > 0$, then the co-explosiveness in x_t migrates to y_t with a lag of i – meaning co-explosive episode in x_t precedes that in y_t . On the other hand, if $i < 0$, then the co-explosive in y_t dominates the co-explosive episode in x_t by i number of periods.

In another case, if $\beta_z > 0$ and $\beta_x = 0$ then it means there is an explosive episode present in y_t which is driven by an unknown process z_t . Given these terms, the hypotheses for co-explosive behaviour or migration can be written as below.

$$H_0 : \beta_x > 0, \beta_z = 0 \quad (\text{Co-explosive})$$

$$H_1 : \beta_x = 0, \beta_z > 0 \quad (\text{Not co-explosive})$$

Under H_0 , x_t and y_t are considered co-explosive as the linear combination $y_t - \mu_y - \beta_x x_{t-i}$ is $I(0)$. Under H_1 , the process is not considered co-explosive, but they differ in the sense the process retains an explosive dynamics which is z_t . If $\beta_x > 0$ and $\beta_z > 0$, then the series x_t cannot alone be used to explain the explosive behaviour in y_t as it depends on z_t . It will instead be treated as part of alternative of no co-explosivity.

Heteroskedasticity for $\varepsilon_{y,t}$ will be allowed under the assumption $\varepsilon_{y,t} = \sigma_t v_t$ where $v_t \sim IID(0, 1)$. σ_t is the volatility term and it satisfies $\sigma_t = w(t/T)$ where $w(\cdot)$ is non-stochastic and strictly positive. This assumption implies that the error variance is non-stochastic and demonstrates that it allows for a number of jumps. The usual homoskedasticity assumption ($\sigma_t = \sigma$) for all t is also allowed here since $w(s) = \sigma$. ADSR makes this assumption only for the purpose of simplicity, while leaving the possibility for relaxation for cases where $w(\cdot)$ is stochastic

and independent of v_t , without affecting the main results, if required. The same approach is adopted for this paper too. The following KPSS statistic is used to test the hypothesis.

$$S = \hat{\sigma}_y^{-2} (T - |i|)^{-2} \sum_{t=i1(i>0)+1}^{T+i1(i<0)} \left(\sum_{s=i1(i>0)+1}^t \hat{e}_{y,s} \right)^2, \quad (3.14)$$

where $\hat{e}_{y,t} = y_t - \hat{\mu}_y - \hat{\beta}_x x_{t-i}$ is the ordinary least square (OLS) residual from the regression y_t on a constant and x_{t-i} . Here $\hat{\sigma}_y^2 = (T - |i|)^{-1} \sum_{t=i1(i>0)+1}^{T+i1(i<0)} \hat{e}_{y,t}^2$. The aim of these general statistics is to identify model errors and to see whether the linear combination between the two series are $I(0)$ or $I(1)$. This is applied to this testing because it is assumed that the series x_t contains an explosive episode. Below is the asymptotic (large sample) behaviour of S for all models (1 – 4) under the assumption of heteroskedasticity ($\varepsilon_{y,t} = \sigma_t v_t$) as given in their paper.

(i) Under H_0 : $S \xrightarrow{w} \int_0^1 V^n(r)^2 dr$ where $V^n(r) = W^n(r) - rW^n(1)$ and,

(ii) Under H_1 : $S = O_p(T^{2\alpha_{z,1}})^{-1}$.

The result of (i) clearly shows that H_0 of S is dependent on the heteroskedasticity that is present in $\varepsilon_{y,t}$. This makes the limit critical values of the KPSS infeasible or inappropriate (due to the obvious heteroskedasticity) as opposed to the case of homoskedasticity, where $W^n(r) = \int_0^r dW(s) = W(r)$. Since in a heteroskedastic case, homoscedastic critical values of KPSS cannot be used to guarantee any large sample size robustness of S , a bootstrap procedure is used.

If the error terms were identically distributed and independent, with a common variance, then a ‘residual bootstrap’ could also be used to make good inferences MacKinnon (2006). However, since the error terms are not identically distributed and not independent, the idea of a ‘residual bootstrap’ is dropped, instead we could go with either the ‘pairs bootstrap’ procedure which was proposed by Freedman (1981) or the ‘wild bootstrap’ procedure which was proposed by Wu (1986). However, since there are no instrumental variables involved in our model, and since we are only interested in residuals and not the data, the wild bootstrap procedure will be employed as in ADSR. The general idea behind the wild bootstrap method is the regressors will remain as sample value, but the response variable will be resampled based on the residual values. According to the model (Wu, 1986) originally proposed, a new y will be computed for each replication based on the following.

$$y_i^* = \hat{y}_i + \varepsilon_i v_i$$

where the residuals are randomly multiplied by v_i which is a random variable having 0 as mean and 1 as variance. Since homoskedastic critical values of KPSS cannot be relied upon to guarantee large sample size robustness of S in a heteroskedastic case, ADSR uses the following algorithm to conduct the bootstrap procedure.

1. Generate bootstrap sample via $y_t^b = w_t \hat{e}_{y,t}$, $t = i1(i > 0) + 1, \dots, T + i1(i < 0)$ where w_t denotes an $IIDN(0, 1)$ sequence.
2. Compute the bootstrap statistic S_b via $\hat{e}_t^b$ which is generated from the regression of y_t^b on the constant and x_{t-i} . i.e.

$$S_b = \hat{\sigma}_b^{-2} (T - |i|)^{-2} \sum_{t=i1(i>0)+1}^{T+i1(i<0)} \left(\sum_{s=i1(i>0)+1}^t \hat{e}_{y,s} \right)^2,$$

$$\text{where } \hat{\sigma}_b^2 = (T - |i|)^{-1} \sum_{t=i1(i>0)+1}^{T+i1(i<0)} \hat{e}_t^{b2}$$

3. Then repeat for $b = 1, 2, 3, \dots, B$ bootstrap replications.
4. Calculate the upper-tail significance level (π) critical value which is c_π^B of the empirical cumulative distribution function of S_b .
5. Finally, if the KPSS statistic S is greater than c_π^B , then the null hypothesis shall be rejected in favour of the alternative hypothesis.

ADSR provides the asymptotic behaviour of S_b in the following way under the assumption of heteroskedasticity for all models (1–4).

(i) Under H_0 : $S_b \xrightarrow{w} p \int_0^1 V^n(r)^2 dr$,

(ii) Under H_1 : $S_b = O_p(T^{\alpha_{x,1}})^{-1}$.

According to the null hypothesis, the limit distribution of S_b is the same as S . That is, when the number of bootstrap replications is large enough, which is our case will be, the empirical cumulative distribution function of S_b will be $\Pr(S > c_\pi^B) = \pi$. This means the size of S will be controlled asymptotically. The reason why we are able to achieve this robustness is because the heteroskedasticity that is present in the original errors $e_{y,t}$ are replicated in $y_t^b = w_t \hat{e}_{y,t}$ (bootstrap data). The limit null distribution of S_b does not involve the series x_t . Hence it is not necessary to include x_{t-i} when we are constructing the bootstrap residuals $\hat{e}_t^b$. The reason is that these residuals can be generated from the regression of y_t^b on a constant alone without including the regressor x_{t-i} . However, in finite samples, when $\alpha_{x,1}$ is close to 1 the effect of x_t might be significant on S . Therefore, it is likely that when we exclude x_{t-i} in the construction of $\hat{e}_t^b$, there is a chance of the finite sample distributions of S and S_b to be less similar compared to how it would have been had we included it. Interestingly, ADSR found this to be the case. They strongly advocate including x_{t-i} in the bootstrap regressions.

Another feature in the asymptotic behaviour of S is that while the ADSR's testing procedure requires the information on x_t , it does not require the procedure to model the specific characteristics of x_t . In short, we do not need to know the form of the explosivity that is present in

the data or which model (1 – 4) exactly generated it. This makes the testing procedure fairly simple and efficient.

Our data are going to have multiple explosive episodes. However, this procedure will still be valid if these bubbles in x_t are formed from sequentially combining the DGPs of all models. Therefore, this procedure can easily be used to test whether a linear combination of x_t and y_t is stationary or not when x_t contains multiple bubbles. Moreover, another advantage of this procedure is that when there is serial dependence in $e_{y,t}$, S_b does not require the long-run variance estimator for its construction as the samples generated y_t^b from the bootstrap procedure does not contain any serial dependence.

We have mentioned the lead/lag parameter i previously, but it must be noted that under the null hypothesis of co-explosivity, this parameter is generally not known. The regression residuals that S and S_b use for its construction is derived from the fitted regression $\hat{e}_{y,t,j} = y_t - \hat{\mu}_y - \hat{\beta}_x x_{t-j}$ [j is user-chosen value for i]. In order to test the possibility that $j \neq i$, ADSR decomposes $\hat{e}_{y,t,j}$ as

$$\hat{e}_{y,t,j} = \beta_x r_{1,t,i,j} + r_{2,t,j}$$

where $r_{1,t,i,j}$ (a residual) is generated from a regression of $x_{t-i} - x_{t-j}$ on a constant and x_{t-j} while $r_{2,t,j}$ (a residual) is generated from a regression of $\varepsilon_{y,t}$ on a constant and x_{t-j} . The residual $r_{1,t,i,j}$ directly contributes to $\hat{e}_{y,t,j}$ if an incorrect choice of j is chosen. If $j = i$ for all t then $r_{1,t,i,j} = 0$. The parameter i is determined by calculating $\hat{\sigma}_{y,j}^{-2} = (T - |j|)^{-1} \sum_{t=j1(j>0)+1}^{T+j1(j<0)} \hat{e}_{y,t,j}^2$ for a range j and by estimating the value of i by the value of j for which $\hat{\sigma}_{y,j}^{-2}$ is minimised. That is, we can estimate the parameter i using $\hat{i} = \operatorname{argmin}_{y \in J} \hat{\sigma}_{y,j}^2$ where J is the set of j values considered and i is an element of J . The finite sample properties including power performance of the wild bootstrap procedure which is based on S is also confirmed by running Monte Carlo simulations which proves to be strong and robust.

As a further investigation method, we will also use Johansen (1991)'s test (Johansen Maximum Likelihood Cointegration test for the pairs that show contemporaneous co-explosive relationships in order to confirm that there is a long-term relationship between them. The results are reported in appendix 4.

Chapter 4

Data Description

The following 12 variables are used for the study in this paper.

Table 4.1: Price Series Descriptions and Sources

Variable	Unit	Description
Crude Oil	(\$/bbl)	Average spot price of Brent, Dubai and WTI, equally weighed
Coal	(\$/mt)	Average of Australian Coal and South African Coal
Natural Gas	(\$/mmbtu)	Average of European Natural Gas
Soybeans	(\$/mt)	U.S Gulf Yellow Soybean (nearest forward)
Wheat	(\$/mt)	U.S, Hard Red Winter (HRW)
Barley	(\$/mt)	U.S, Winnipeg Commodity Exchange (wholesale farmers' spot)
Maize	(\$/mt)	U.S, Free on Board (FOB) US Gulf ports price
Sorghum	(\$/mt)	U.S, Free on Board (FOB) US Gulf ports price
Beef	(\$/kg)	Beef (AU/NZ), Free on Board (FOB) port of entry price
Chicken	(\$/kg)	U.S, Georgia Dock weighted average wholesale price
Sugar	(\$/kg)	International Sugar Agreement (ISA) daily price
Coffee	(\$/kg)	International Coffee Organization (ICO) indicator price

The data consist of 600 observations (1970:9 – 2020:8) for all the series of energy (Crude Oil, Coal, Natural Gas) and for all series of non-energy (Soybeans, Wheat, Barley, Maize, Sorghum, Beef, Chicken, Sugar, Coffee) except for Corn, for which only 368 observations are available (1990:1 – 2020:8). All the price series are downloaded from World Bank as monthly nominal figures and deflated using US Consumer Price Index (CPI) series from Federal Reserve Bank of St. Louis.

Table 4.2: Summary Statistics

Variable	Start Date	End Date	Date (max)	Date (min)	DF	Δ DF
Crude Oil	Sept 1970	Aug 2020	Jul 2008	Dec 1970	−3.430***	−12.268***
Coal	Sept 1970	Aug 2020	Jul 2008	Aug 2002	−2.548	−10.402***
Natural Gas	Sept 1970	Aug 2020	Nov 2008	May 2020	−2.827*	−10.968***
Soybeans	Sept 1970	Aug 2020	June 1973	May 2001	−2.037	−12.175***
Wheat	Sept 1970	Aug 2020	Feb 1974	Dec 2016	−2.040	−11.458***
Barley	Sept 1970	Aug 2020	Feb 1974	Aug 2020	−2.350	−12.956***
Maize	Sept 1970	Aug 2020	Oct 1974	Jul 2000	−2.011	−11.797***
Sorghum	Sept 1970	Aug 2020	Oct 1974	Jul 2000	−2.161	−11.955***
Beef	Sept 1970	Aug 2020	Aug 1973	Oct 1998	−2.034	−13.332***
Chicken	Sept 1970	Aug 2020	Jul 1973	Apr 2020	−2.833*	−14.567***
Sugar	Sept 1970	Aug 2020	Nov 1974	June 1985	−2.542	−11.224***
Coffee	Sept 1970	Aug 2020	Apr 1977	Aug 2002	−2.236	−11.500***

*, **, *** denote the significance level at the 10%, 5% and 1% levels respectively. The frequency of the data is monthly and sample size is 600 for all series.

There are notable differences between the behaviour of energy market and the non-energy market, and important observations can be made from the data we are examining. The energy sector seems to have reached its peak in mid-2008. From 1980s till 2003, the energy prices remained comparatively low. However, starting 2003, a consistent rise in the energy market can be observed, whose rise can be attributed to many factors starting from the rise of industrialising China, the tension in the Middle East due to wars and conflicts, falling dollar value during that period and also the financial speculation. The lowest price for Natural Gas can be attributed to the start of the pandemic in the beginning of 2020 and the low Coal prices in August 2002 can be linked to the world economic recession of 2001. Crude oil hit its lowest in Dec 1970, however, this also coincides with the period our sample started (Sep 1970), therefore this cannot be linked to any particular event. Similarly, it can be found from the data that the prices of major agricultural commodities including Soybeans, Wheat, Barley, Maize, Sorghum, Beef, Chicken and Sugar also reached its high in the early 1970s (between June 1973 and 1974). This can be mainly attributed to the stimulated world demand during that time including Soviet Union and several other centrally planned countries' sudden demand for grain in the world market. The peak in coffee prices in 1977 is due to the major frost in the largest coffee producing country, Brazil, in the year 1975 which led to a shortage of coffee rising price in the following years. The peak of corn prices in 2012 is due to drought and heatwave the US (largest producer of corn) experienced in that year which drove the production of corn down significantly. On the other hand, the record level drop in price of Soybeans, Maize, Sorghum, Coffee and Corn can be linked to the Asian Financial Crisis that started in the year 1997. The lowest record prices for Barley and Chicken in the year 2020 is due to the Covid-19 pandemic driven by weak global

demand, especially in Asia.

In the next section, we will look at the results we have generated from using the PSY test and the ADSR test on the energy and non-energy prices and discuss the likely originations of the bubbles and bubble migrations.

Chapter 5

Results

5.1 Explosive autoregression in the individual series

Before we apply the co-bubbling test to the price series, explosive autoregressive regimes to be present in the individual series is a pre-condition for co-explosivity using the ADSR. The PSY procedure we detailed in the methodology part is employed to detect any explosive autoregressive regimes in the price series, for which the results are provided in table 3. In this empirical application, (Phillips, et al. (2015a, b)’s recommended rule of thumb $r_0 = 0.01 + 1.8/\sqrt{T}$ is used in setting the minimum window size. The minimum window size should be large enough to allow the initial estimation, but not also too large in order to avoid missing short periods of exuberance. Regarding the autoregressive lag length, according to Vasilopoulos, et al. (2020), the simulation evidence suggests the right-tailed methodologies that we described above work well when the number of lags are fixed at a small value. When the lag selection is determined by Information Criteria such as Bayesian Information Criteria (BIC), it can result in severe size distortions. These right-tailed unit root tests also require limit distributions of SADF, GSAD and BSADF (Vasilopoulos, et al., 2020). While these distributions are non-standard, they depend on the size of the window period. Taking the above into consideration, a minimum window size of 50 and a maximum lag length of 6 is selected to generate the standard ADF, SADF (PWY, 2011) and GSADF (PSY, 2015). Same as ADSR, to assess the statistical significance of the statistics that are computed, we also use critical values that allow for heteroskedasticity by employing a wild bootstrap with 499 replications and also use finite sample critical values for 600 observations from Vasilopoulos, et al. (2020) for the purpose of completeness.

Table 5.1: Result of tests for explosive autoregressive in individual series of energy and non-energy sector

	ADF	SADF	GSADF
<i>Energy Sector</i>			
Crude Oil	−2.6256	0.2679	6.0946 ^{1,a}
Coal	−2.8408	1.6383 ^{3,b}	2.6749 ^a
Natural Gas	−2.4464	−0.1118	2.5339 ^b
<i>Non-Energy Sector</i>			
Soybeans	−2.4036	−1.691	2.2447 ^b
Wheat	−2.0769	−1.2963	2.9703 ^a
Barley	−2.6643	−0.9747	2.3764 ^b
Maize	−1.9263	−0.9866	2.6837 ^{3,a}
Sorghum	−1.818*	−0.6914*	1.8382*
Beef	−2.2288*	−1.3774*	1.6592*
Chicken	−2.4241	1.4016 ^c	3.7767 ^{1,a}
Sugar	−3.6257	1.8192	3.0334 ^c
Coffee	−2.343	3.2767 ^{1,a}	3.7821 ^{2,a}

Note: The superscripts 1, 2, 3 indicate the rejection of the null hypothesis at the significance levels 1%, 5% and 10% respectively using the right-tail Monte Carlo critical values based on wild bootstrap with 499 replications whereas superscripts a, b, c indicate the rejection at the significance levels 1%, 5% and 10% respectively using right-tail tabulated critical values from Vasilopoulos, et al. (2020).

As can be seen from table 1, the rejections for the null hypothesis of unit root are obtained at the conventional significance levels for all energy sector and for seven of the series in the non-energy sector from the PSY test, when the right-tailed tabulated critical values from Vasilopoulos, et al. (2020) are used. GSADF statistics are significant at 0.01 level for six of the series when the tabulated critical values are used for rejection. However, when we use wild bootstrap critical values, the result (explosive behaviour) becomes a little weaker as happened in ADSR. Only two series (Crude Oil, Chicken) are found to be having explosive behaviour at the 0.01 level using wild bootstrap critical values. This shows that the rejections are fewer and weaker when we account for heteroskedasticity in our data i.e. using wild bootstrap critical values, when testing for explosive behaviour using PSY tests. This is consistent with the results obtained by ADSR. Having said that, we find evidence of explosive autoregressive behaviour consistent with the presence of a bubble in ten out of the twelve series using the PSY test.

The SADF test is consistent only when we are dealing with single episode boom-busts. While GSADF which is an extension of SADF, holds the same hypothesis, it covers a larger number of subsamples has greater flexibility and substantial power gains as mentioned earlier, and it better suited to detect multiple bubbles. Therefore, only GSADF test results are considered for the purpose of detecting explosivity in our time series.

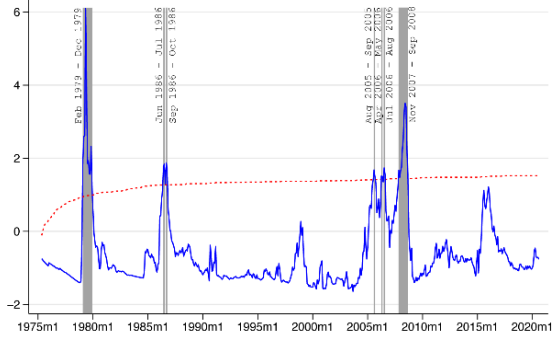
Subsamples are also employed for the two series (Sorghum, Beef) for which rejections were not obtained based on the full sample for GSADF and SADF statistics, for the period 2006M1 – 2010M12 and 1990M1 – 2000M12 respectively. These subsamples were selected based on the largest visible fluctuations seen from the graphs. The subsamples with a maximum lag of five and eight resulted in a rejection for both the series, making all the series we have introduced so far explosive autoregressive regimes. The results are given in table 4.

Table 5.2: Result of tests for explosive autoregressive in individual series (sub-sample)

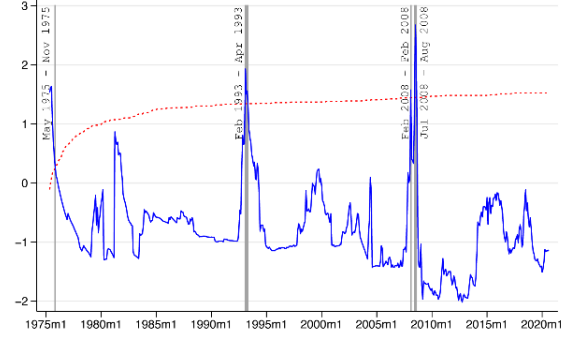
	ADF	SADF	GSADF
Sorghum	−2.8098	0.1739	1.6110 ^c
Beef	−1.3616	0.8108	1.9491 ^c

Note for table 3 applies here.

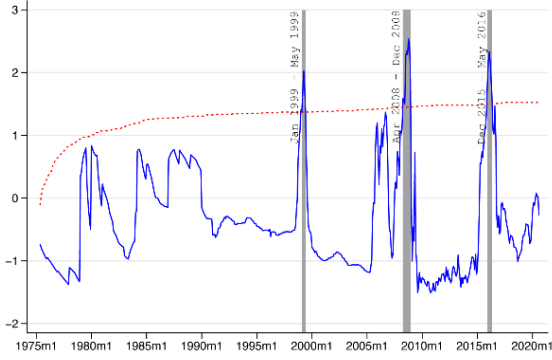
While the PSY test proposes using the market fundamentals to determine whether there is a rational bubble or not in a series, we will not be using any market fundamentals for any of the series due to the fact that there are no proper fundamentals available for the commodity price series we have selected. We interpret the results here while keeping in mind that there is a possibility that any or some explosiveness we have identified in the series may be generated by the fundamentals. The estimated PSY bubble origination and termination dates for each episode are given in appendix 3, and it is also visually shown in figure 4.



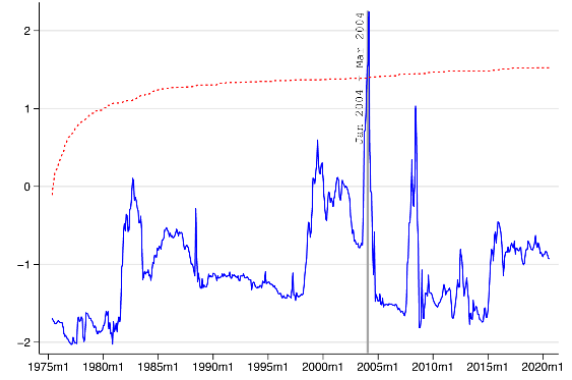
(a) Crude Oil



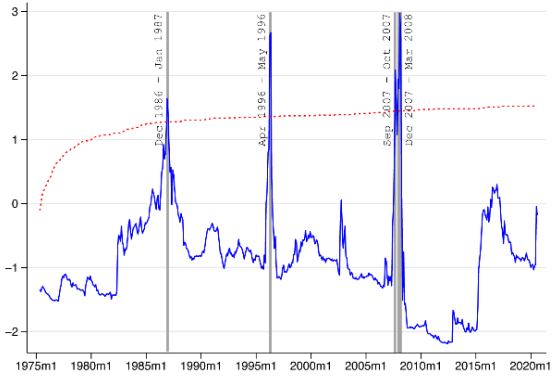
(b) Coal



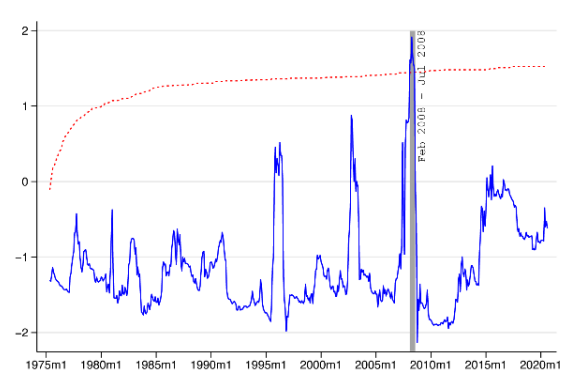
(c) Natural Gas



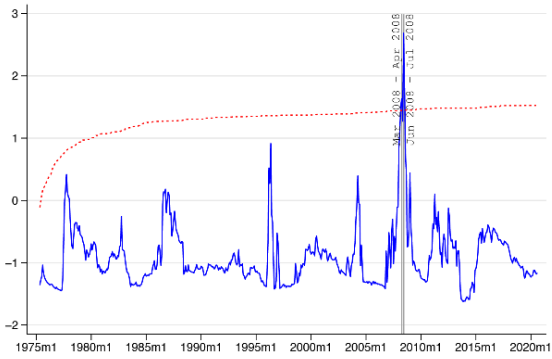
(d) Soybeans



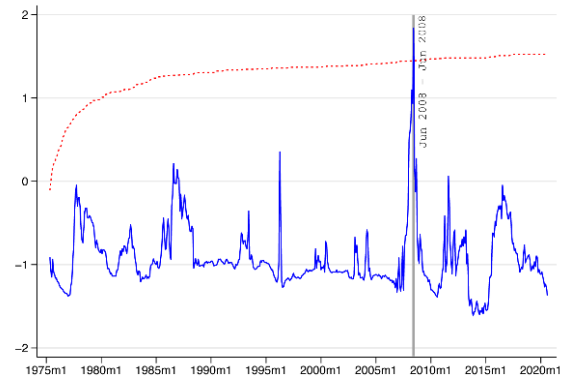
(e) Wheat



(f) Barley



(g) Maize



(h) Sorghum

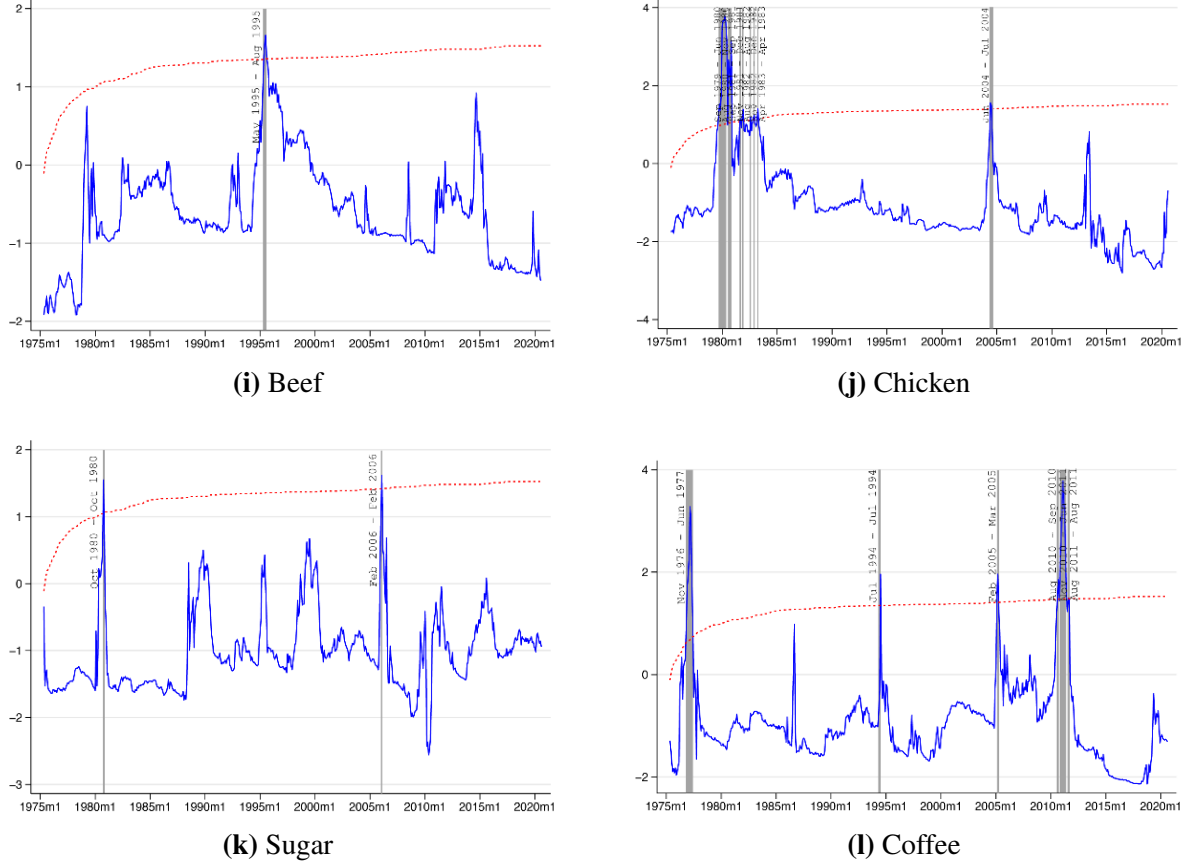


Figure 5.1: Date-stamping of the explosive bubbles in the individual price series

Our study does not use any market fundamentals to confirm the explosivity in the individual time series due to the unavailability of proper proxies for the price series we have used in the study. However, as ADSR mentioned in their paper, the test we are using is informative enough to be used in applications where fundamental data are not available too. Because the test seeks evidence against the co-explosive behaviour which is the null hypothesis of our test. The fundamental data's absence does not affect the ability of the test to reject this null hypothesis when it is false. As this is confirmed in ADSR paper also, we are following the same procedure.

5.2 Test for co-explosive behaviour

The lags and lead parameters that are used to construct the residuals to generate the S and S_b statistics are estimated by using $\hat{i}$ with $J = \{-24, -23 \dots 0, 23, 24\}$. The same procedure employed by ADSR and as explained in the methodology, is used to generate 5000 bootstrap replications. Serial correlation in $\varepsilon_{y,t}$ is accounted for by replacing the short-run variance estimator $\hat{\sigma}_y^2$ with Newey & West (1994) suggested long-run estimator which employs a quadratic spectral kernel with automatic lag selection. Although Monte Carlo simulations suggest size distortions, it also shows that this procedure performs adequately well. They also prove that their procedure is asymptotically equivalent to one that is optimal under a mean-squared error

loss function. Co-bubble results for all possible pairs are generated except for Crude Oil as the results do not show any explosive regimes in them. The possibility of Crude Oil being y_t and x_t is considered in the co-explosive behaviour test. As in ADSR, the results for our test is also provided for the series for which $p \geq 0.025$ regardless of which series were specified as y_t and x_t . The graphs for all the pairs for which co-explosivity is found are given under Figure 5.

From Appendix 5, we can see that there is evidence of co-explosivity in the energy sector involving Coal and Natural Gas, and in the non-energy sector involving all price series. While there is co-explosive behaviour between Coal and Natural Gas in the energy sector, evidence of co-explosivity is found for 10 cases between non-energy sector series: Soybeans – Wheat, Soybeans – Maize, Soybeans – Sorghum, Soybeans – Beef, Soybeans – Coffee, Wheat – Maize, Wheat – Sorghum, Maize – Sorghum, Maize – Chicken and Chicken – Sugar. There are also four instances of co-explosivity between energy and non-energy sector which are Coal – Natural Gas, Coal – Barley, Coal – Sugar and Gas – Barley.

In the case of the energy sector, for the co-explosive pair Coal – Natural Gas, the $\hat{i} = -3$ which indicates that Coal prices lead Natural Gas by three months. The result however shows that the Natural Gas on Coal is more significant ($p = 0.122$) than the influence of Coal on Natural Gas ($p = 0.064$). While there is little literature on the relationship between Coal price and Natural Gas price, our results are mainly consistent with the study done by Li, et al. (2017) where they also found that the influence of Natural Gas price on Coal is very strong while the influence of Coal price on Natural Gas price is tiny in the European market. The data we are using is also collected from the European market. This explosive behaviour can be linked to the factors we discussed earlier relating to the industrialisation of China, tension in the Middle East and the financial speculation.

In the non-energy sector, for the pairs Soybeans – Maize, Soybeans – Coffee, Wheat – Maize, Wheat – Sorghum and Maize – Sorghum, $\hat{i} = 0$ which suggests contemporaneous co-explosive relationships between them. There is no known literature on the exact relationship between these commodities per se. However, it must be noted that all these commodities fall under the agriculture category under non-energy sector according to the World Bank classification which explains the likely or similar nature of the products. As can be seen from the graphs also, the movements are very similar across all these commodities, mainly for the reason that the same technology and techniques are used for the production of these crops. Moreover, these commodities can be used a complementary product in many countries which might also be a reason for their similar trends. We ran the Johansen Cointegration to investigate this further. According to the results, there is a long-run relationship between at least three variables, which confirms that three of these price series exhibit a cointegration relationship at 1% significance level, which adds weight to what we just mentioned in the above.

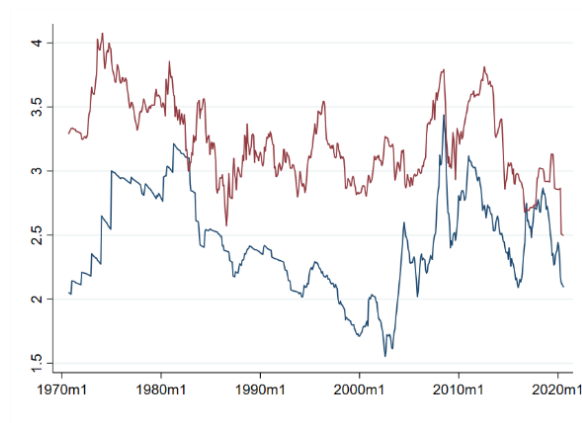
We also find co-explosive behaviour for Soybean – Wheat with $\hat{i} = -6$ indicating that the explosive behaviour in Soybeans price leads Wheat price by six months. This is an interesting observation because the data we have used for this pair are from the US. The Southeastern

farmers in the US started double-cropping Soybean and Wheat in 1970s according to (Marra & Carlson, 1986). This is the practice of raising two crops per year in the same field and this has been a common practice in many regions of the US since then, which might explain the co-explosive behaviour between them. Another interesting observation we make is the result for Beef – Soybeans with $\hat{i} = -3$ indicating explosive behaviour in the price of Beef leads explosive behaviour in Soybeans by three months. Soybeans is a major protein source for cattle. Therefore, any increase in the global price of beef or global demand is likely going to lead to an increase in demand for Soybeans as the producers would want to produce more and stockpile Soybeans for that matter. For Maize – Chicken pair, $\hat{i} = -22$ indicates a long-term relationship between Maize and Chicken where it suggests a co-explosive relationship between them but the impact period is estimated to be roughly 22 months. According to Andam, et al. (2017), Maize accounts for poultry feed and its availability and price have huge implications for the poultry production. However, it might not be rational to attribute the relationship to the factors of input for poultry without further research given the long time gap between them. It adds weight to the aforementioned statement when we look at the Chicken – Sugar relationship. The $\hat{i} = -17$ for that pair, but there is no known relationship between Chicken and Sugar.

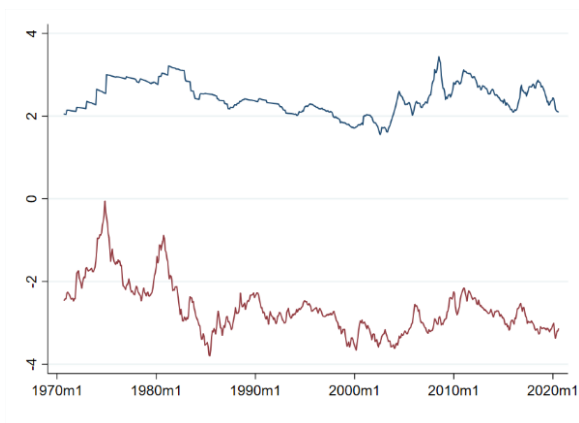
Now we look at the co-explosive behaviour between the energy and non-energy sector. The Coal – Barley pair with $\hat{i} = 5$ means the explosive regime in Barley is migrating to Coal after five months. A similar pattern is found in the Coal – Sugar and Gas – Barley pairs. In the former, the explosive episode in Sugar migrates to Coal after 15 months while in the latter, it takes seven months for the explosive bubble in Barley to reach Natural Gas. Again, in the case of the former, we can see from the bubble stamping graphs also there was an explosive episode in Sugar in February 2006 and a similar episode occurs in Coal in the beginning of 2008. When we look at the date stamping graphs of Barley and Natural Gas, we can see that there was an explosive episode in Barley in February 2008 and Natural Gas experienced an explosive episode between April 2008 and December 2008. This pattern is visible for all the three pairs that involve energy and non-energy sector; hence we could say that biofuels might be contributing to this relationship and/or transfer of explosivity between these two sectors. Hence our study gives weight to the theories Nicola, et al. (2016) and Peñaranda & Micola (2009) have provided in their papers on the relationship between energy and non-energy sector with biofuels being a possible determinant. Biofuels production can be enhanced with Natural Gas (blending Natural Gas with biofuels) according to Wright, et al. (2015). With further research into this area, this theory can be confirmed.



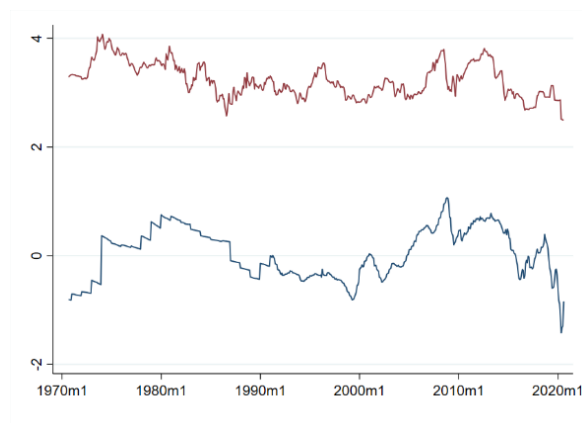
(a) (i) Coal, (ii) Natural Gas



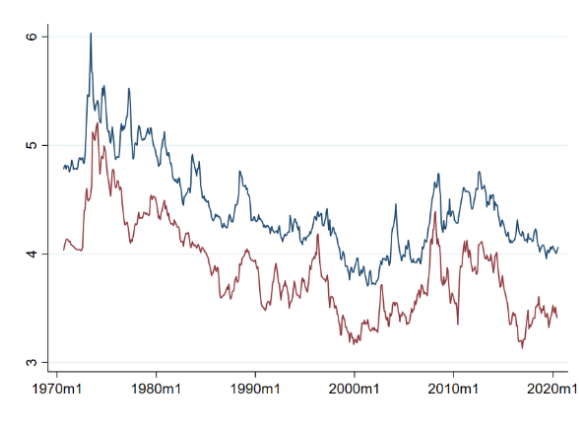
(b) (i) Coal, (ii) Barley



(c) (i) Coal, (ii) Sugar



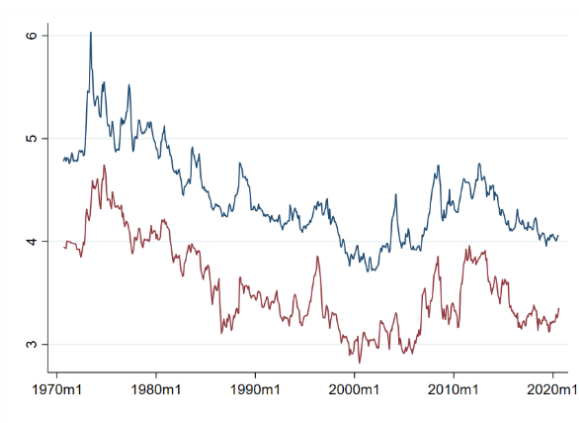
(d) (i) Natural Gas, (ii) Barley



(e) (i) Soybeans, (ii) Wheat



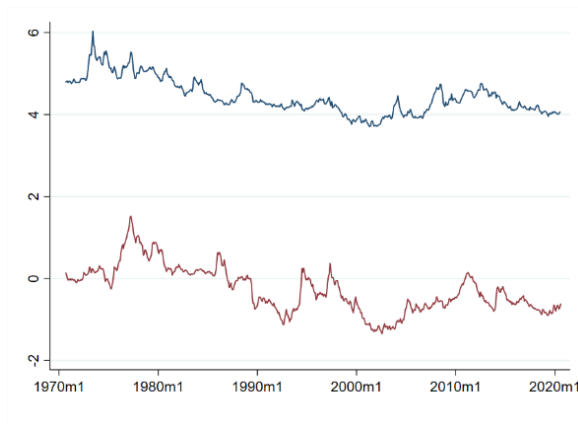
(f) (i) Soybeans, (ii) Maize



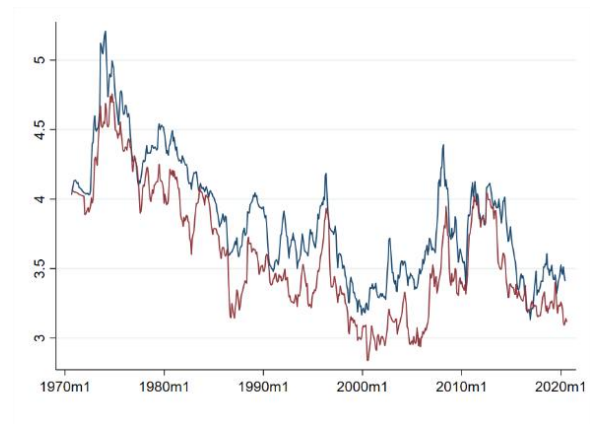
(g) (i) Soybeans, (ii) Sorghum



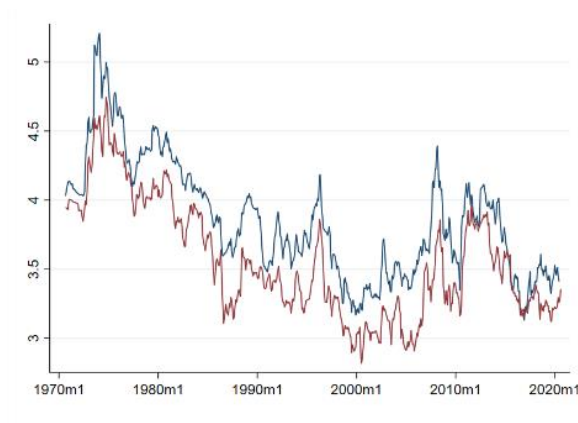
(h) (i) Soybeans, (ii) Sorghum



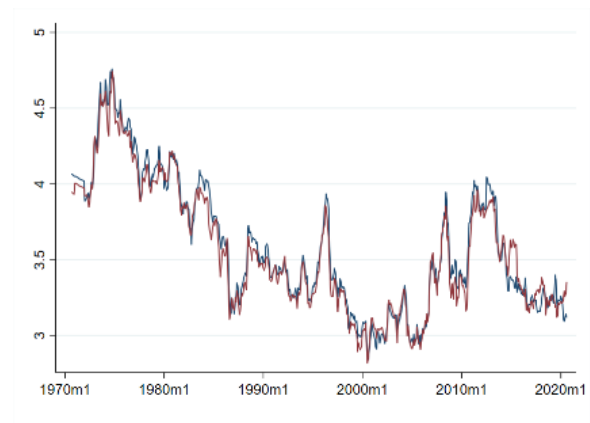
(i) (i) Soybeans, (ii) Coffee



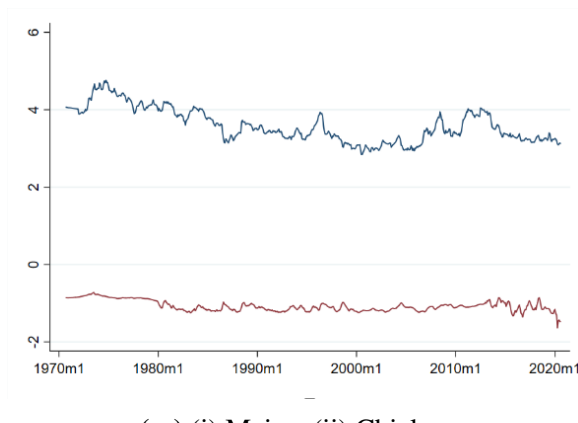
(j) (i) Wheat, (ii) Maize



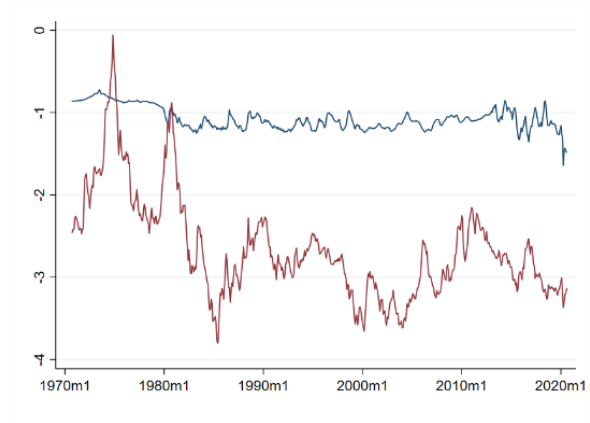
(k) (i) Wheat, (ii) Sorghum



(l) (i) Maize, (ii) Sorghum



(m) (i) Maize, (ii) Chicken



(n) (i) Chicken, (ii) Sugar

Figure 5.2: Price series (pairs) with co-bubbling evidence: Series (i) prices: Series (ii) prices:

5.3 Policy Implications

Price volatility in the energy and non-energy sector has huge implications both economically and socially for nations, especially the developing or less developed countries. What we have shown in this paper are the most commonly demanded raw materials in the energy and non-

energy sector, their explosive price behaviour and how price bubbles might migrate from one sector to the other. This study shows the time period where the price series have departed from their possible fundamental values (explosive bubbles). It also draws a possible relationship with other price series and estimates a time period in months on the likelihood of the effects in one series migrating to the other. Policymakers can make use of these studies to narrow down their focus on the commodities that most need attention and prioritise the markets which display bubbles. Policies can be designed to mitigate possible price exuberances and bubble migrations such as Coal – Barley and Wheat – Maize which show the strongest co-bubbling results. Economies that highly depend on other countries for energy and agricultural products such as Maldives can make use of this study and similar studies to more efficiently plan and manage their buffer stock and improve energy and food security in a way that it does not cause unexpected price surges in their fragile economies. The level of explosivity in the price series the PSY identifies and the time period for migration between these price series that the ADSR forecasts can be a vital source of information for the national and state governments around the world.

Chapter 6

Concluding Remarks

In this paper we presented the likely factors that contribute to the price surges and bubbles in the energy and non-energy sector and drew conclusions on the possible relationships between the price series using PSY and ADSR. Before we employed ADSR to study possible bubble migrations between the price series, we tested all the price series using PSY test to confirm that there is at least one explosive episode in each series. All the 12 series we used are found to have explosiveness in them. The ADSR technique we used in this paper uses a wild-bootstrap procedure that generates critical values that are robust to heteroskedasticity. Results of the Monte Carlo simulations presented in this recently developed ADSR technique confirm good finite sample size and power performance.

The date-stamping of bubbles via PSY shows a pattern of the estimated bubble episodes coinciding with major events that took place in the last 50 years. The application of ADSR confirms 14 co-bubbling pairs (out of the 66 possible pairs of price series) of series of which five pairs (Soybeans/Maize, Soybeans/Coffee, Wheat/Maize, Wheat/Sorghum and Maize/Sorghum) show contemporaneous co-explosive behaviour. Johansen Maximum Likelihood Cointegration test for these pairs confirm at least three cointegrating relationships among them. ADSR also shows bubble migrations taking place between the energy and non-energy sector (e.g. Natural Gas and Barley) and also within each sector. In the case of the former it can be linked to the biofuel initiatives and programmes that are taking place around the globe and in the latter, it can be linked to the production techniques and several other factors as well.

We also found some pairs for which we were not able to accurately identify a possible reason for their migrations such as Chicken and Sugar which shows a 15-month gap in the migration. There are several factors which might be causing this strange price behaviour for which more data might be required.

In light of the evidence we have presented in this paper, future studies can be conducted on these specific price series by focusing on the ingredients and factors of production that are used for each commodity in order to identify the reason why there might be a relationship between them (Chicken and Sugar for example) and use similar econometric techniques we have used in this study. Moreover, policymakers can make use of studies like these to ensure food security

and mitigate price fluctuations, especially countries which are import dependent such as the Maldives.

Bibliography

- Baffes, J. (2007). Oil spills on other commodities. *Policy Research Working Paper 4333*, pages 1–23.
- De Nicola, F., De Pace, P., and Hernandez, M. A. (2016). Co-movement of major energy, agricultural, and food commodity price returns: A time-series assessment. *Energy Economics*, 57:28–41.
- Gordon, M. J. and Shapiro, E. (1956). Capital equipment analysis: The required rate of profit. *Management Science*, 3(1):102–110.
- Gronwald, M. (2016). Explosive oil prices. *Energy Economics*, 60:1–5.
- Gutierrez, L. (2013). Speculative bubbles in agricultural commodity markets. *European Review of Agricultural Economics*, 40(2):217–238.
- Gürkaynak, R. S. (2008). Econometric tests of asset price bubbles: Taking stock. *Journal of Economic Surveys*, 22(1):166–186.
- Kortian, T. (1995). Modern approaches to asset price formation: A survey of recent theoretical literature. *Research Discussion Paper 9501*, pages 1–34.
- Peñaranda, F. and Micola, A. R. (2009). The emergence of biofuels and the co-movement between crude oil and agricultural prices. Barcelona, Spain. Department of Economics and Business, Universitat Pompeu Fabra.
- Phillips, P. C. B., Wu, Y., and Yu, J. (2011). Explosive behavior in the 1990s nasdaq: When did exuberance escalate asset values? *International Economic Review*, 52(1):201–226.
- Roy, S. and Kemme, D. M. (2020). The run-up to the global financial crisis: A longer historical view of financial liberalization, capital inflows, and asset bubbles. *International Review of Financial Analysis*, 69.
- Saghaian, S. H. (2010). The impact of the oil sector on commodity prices: Correlation or causation? *Journal of Agricultural and Applied Economics*, 42(3):477–485.

- Saitone, T. L. and Sexton, R. J. (2007). Alpaca lies? speculative bubbles in agriculture: Why they happen and how to recognize them. *Review of Agricultural Economics*, 29(2):286–305.
- West, K. D. (1987). A specification test for speculative bubbles. *The Quarterly Journal of Economics*, 102(3):553–580.

Appendices

Appendix 1 – Right-tail ADF statistics

Price Series	Right-tailed	Statistic	RTMC90	RTMC95	RTMC99	Tab90	Tab95	Tab99
Crude Oil	ADF0	−2.6256	−0.4439	−0.0577	0.6523	−0.3842	−0.0146	0.6392
	SADF	0.2679	1.4912	2.0241	2.9011	1.2435	1.525	2.0712
	GSADF	6.0946	2.8914	3.2394	3.8844	1.9873	2.237	2.6413
Coal	ADF0	−2.8408	−0.3675	−0.0137	0.4926	−0.3842	−0.0146	0.6392
	SADF	1.6383	1.5341	2.0265	2.6142	1.2435	1.525	2.0712
	GSADF	2.6749	3.0606	3.4209	4.9571	1.9873	2.237	2.6413
Natural Gas	ADF0	−2.4464	−0.5819	−0.1507	0.7	−0.3842	−0.0146	0.6392
	SADF	−0.1118	1.5468	2.0448	2.5889	1.2435	1.525	2.0712
	GSADF	2.5339	3.2046	3.5374	4.1081	1.9873	2.237	2.6413
Soybeans	ADF0	−2.4036	−0.3332	0.0483	0.8072	−0.3842	−0.0146	0.6392
	SADF	−1.691	1.3002	1.6907	2.6989	1.2435	1.525	2.0712
	GSADF	2.2447	3.1227	3.4293	4.326	1.9873	2.237	2.6413
Wheat	ADF0	−2.0769	−0.4596	−0.1285	0.6546	−0.3842	−0.0146	0.6392
	SADF	−1.2963	1.3623	1.8348	2.1704	1.2435	1.525	2.0712
	GSADF	2.9703	3.1127	3.5832	4.4603	1.9873	2.237	2.6413
Barley	ADF0	−2.7862	−0.3719	0.0402	0.943	−0.3842	−0.0146	0.6392
	SADF	−1.1372	1.3567	1.8499	2.5343	1.2435	1.525	2.0712
	GSADF	1.9134	2.9648	3.2653	3.9369	1.9873	2.237	2.6413
Maize	ADF0	−1.9263	−0.3639	−0.0768	0.5763	−0.3842	−0.0146	0.6392
	SADF	−0.9866	1.3963	1.6723	2.2515	1.2435	1.525	2.0712
	GSADF	2.6837	2.7451	3.0962	3.9659	1.9873	2.237	2.6413
Sorghum	ADF0	−1.818	−0.46	−0.1862	0.483	−0.3842	−0.0146	0.6392
	SADF	−0.6914	1.4573	1.7672	2.5617	1.2435	1.525	2.0712
	GSADF	1.8382	2.9626	3.3622	4.0058	1.9873	2.237	2.6413
Beef	ADF0	−2.2288	−0.3302	−0.044	0.8598	−0.3842	−0.0146	0.6392
	SADF	−1.3774	1.2925	1.6848	2.5359	1.2435	1.525	2.0712
	GSADF	1.6592	3.0091	3.3634	4.0719	1.9873	2.237	2.6413
Chicken	ADF0	−2.4241	−0.498	−0.1472	0.4657	−0.3842	−0.0146	0.6392
	SADF	1.4016	1.4137	1.9279	2.6604	1.2435	1.525	2.0712
	GSADF	3.7767	2.8301	3.0947	3.7142	1.9873	2.237	2.6413
Sugar	ADF0	−3.4943	−0.4011	0.0405	1.0041	−0.3842	−0.0146	0.6392
	SADF	−0.3452	1.2605	1.6001	2.35	1.2435	1.525	2.0712
	GSADF	1.6156	3.2614	3.8301	5.0801	1.9873	2.237	2.6413

Appendix 2 – Right-tail ADF statistics (sub-sample)

Price Series	Right-tailed	Statistic	RTMC90	RTMC95	RTMC99	Tab90	Tab95	Tab99
Crude Oil	ADF0	−2.6256	−0.4439	−0.0577	0.6523	−0.3842	−0.0146	0.6392
	SADF	0.2679	1.4912	2.0241	2.9011	1.2435	1.525	2.0712
	GSADF	6.0946	2.8914	3.2394	3.8844	1.9873	2.237	2.6413
Coal	ADF0	−2.8408	−0.3675	−0.0137	0.4926	−0.3842	−0.0146	0.6392
	SADF	1.6383	1.5341	2.0265	2.6142	1.2435	1.525	2.0712
	GSADF	2.6749	3.0606	3.4209	4.9571	1.9873	2.237	2.6413

Appendix 3 – Date-stamps

Series	Episode #	Start	End	Series	Episode #	Start	End
Crude Oil	Episode 1	Feb 1979	Dec 1979	Maize	Episode 1	Mar 2008	Apr 2008
	Episode 2	Jun 1986	Jul 1986		Episode 2	Jun 2008	Jul 2008
	Episode 3	Sep 1986	Oct 1986	Sorghum	Episode 1	Jun 2008	Jun 2008
	Episode 4	Aug 2005	Sep 2005	Beef	Episode 1	May 1995	Aug 1995
	Episode 5	Apr 2006	May 2006	Sugar	Episode 1	Oct 1980	Oct 1980
	Episode 6	Jul 2006	Aug 2006		Episode 2	Feb 2006	Feb 2006
	Episode 7	Nov 2007	Sep 2008		Episode 1	Sep 1979	Jun 1980
Coal	Episode 1	May 1975	Nov 1975		Episode 2	Aug 1980	Nov 1980
	Episode 2	Feb 1993	Apr 1993		Episode 3	Sep 1981	Sep 1981
	Episode 3	Feb 2008	Feb 2008	Chicken	Episode 4	Nov 1981	Dec 1981
	Episode 4	Jul 2008	Aug 2008		Episode 5	Aug 1982	Aug 1982
	Episode 1	Jan 1999	May 1999		Episode 6	Nov 1982	Dec 1982
Natural Gas	Episode 2	Apr 2008	Dec 2008		Episode 7	Apr 1983	Apr 1983
	Episode 3	Dec 2015	May 2016		Episode 8	Jun 2004	Jul 2004
Soybeans	Episode 1	Jan 2004	Mar 2004		Episode 1	Nov 1976	Jun 1977
	Episode 1	Dec 1986	Jan 1987		Episode 2	Jul 1994	Jul 1994
	Episode 2	Apr 1996	May 1996	Coffee	Episode 3	Feb 2005	Mar 2005
Wheat	Episode 3	Sep 2007	Oct 2007		Episode 4	Aug 2010	Sep 2010
	Episode 4	Dec 2007	Mar 2008		Episode 5	Nov 2010	Jun 2011
Barley	Episode 1	Feb 2008	Jul 2008		Episode 6	Aug 2011	Aug 2011

Appendix 4 – Results of Johansen cointegration test statistics

rank	Trace Test		Maximum Eigenvalue Test	
	Trace statistic	Critical Value 1%	Max Eigenvalue Statistic	Critical Value 1%
0	169.7858	76.07	73.3143	38.77
1	96.4715	54.46	44.9383	32.24
2	51.5332	35.65	34.7621	25.52
3	16.7711*	20.04	14.0911*	18.63
4	2.68	6.65	2.68	6.65

Appendix 5 – Results ADSR for co-explosive behaviour

y_t	x_t	$\hat{i}$	P-value
Coal	Gas	−3	0.064
Gas	Coal	+3	0.122
Coal	Barley	+5	0.318
Barley	Coal	−6	0.104
Coal	Sugar	+15	0.224
Sugar	Coal	−14	0.052
Gas	Barley	+7	0.096
Barley	Gas	−7	0.048
Soy	Wheat	−6	0.028
Wheat	Soybeans	+6	0.182
Soybeans	Maize	0	0.092
Maize	Soybeans	0	0.516
Soybeans	Sorghum	−1	0.03
Sorghum	Soybeans	+1	0.394
Soybeans	Beef	+3	0.136
Beef	Soybeans	−3	0.038
Soybeans	Coffee	0	0.07
Coffee	Soybeans	0	0.096
Wheat	Maize	0	0.338
Maize	Wheat	0	0.112
Wheat	Sorghum	0	0.168
Sorghum	Wheat	0	0.094
Maize	Sorghum	0	0.260
Sorghum	Maize	0	0.362
Maize	Chicken	+22	0.032
Chicken	Maize	−22	0.290
Chicken	Sugar	−17	0.104
Sugar	Chicken	+17	0.044